

Generative AI in Higher Education: the Good, the Bad, and the Ugly

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Abstract: We attempt to offer a balanced perspective on the use of Generative AI in education, including the good, bad, and ugly aspects

Keywords: Artificial Intelligence, GenAI, fabricated literature reviews, fake content, plagiarism, misrepresentation of GenAI output, GenAI in computer programming, self-efficacy, cognitive decline

Use of GenAI is often championed by online and traditional, brick-and-mortar educational institutions which laud the potential value to education while adapting curriculum and infrastructure to capitalize on perceived economies. There is also no shortage of research studies that deal with the potential pedagogical advantage of GenAI and peer technology innovations in higher education. This is not to ignore considerable concern in education over the overuse, misuse and abuse of GenAI. But what is missing in the messaging is a concise contextual balance – an overview, if you will, that emphasizes the continuum of advantages, disadvantages and serious problems attendant to the use of GenAI in higher education. This is what we attempt to do in this paper. We offer four exemplars of each category. These exemplars are not intended to be exhaustive. Neither are they unbiased. Nor is there anything significant about the number four. We chose four exemplars because that seemed to strike a reasonable balance between scope and brevity. These exemplars reflect our experience as lifelong educators. Other experiences will differ. But if we are successful, the value of this paper will provide useful starting points for further inquiries along the continuum for others to follow. We begin with the good side of the use of GenAI in higher education.

GOOD #1: Enhancing the Learning Experience

Learning is a foundation of the higher education enterprise, making the impact of AI on learning also an impact on the foundations of higher education. Widely acknowledged as disruptive to current practices, if wisely integrated this impact is expected to ultimately benefit the goal of learning [1]. The [old-fashioned misconception](#) of students as largely identical, passive receptacles for knowledge is one extreme of a spectrum, and AI can be an important factor in moving the needle toward the student as a unique and active learner.

Where in times long past a student could exercise just a little individuality by deciding what to look up at the library, and more recently by using general web searches or Wikipedia, now they can request individualized, custom articles about anything, generated in real time by an AI chatbot, with the desired depth, length, reading level and language, or even a [hybrid of languages](#) [2]. But generating custom articles in real time is just a start.

Socratic questioning has existed for over two thousand years: teach by posing a question to the student that challenges their thinking, then follow up with another. The recent use of generative AI demonstrates that the Socratic method is well-suited to modern AI chatbots. Widely available dedicated commercial versions are [not necessarily highly rated](#), at least not yet. However, general-purpose AI chatbots can work well, so students can prompt the same AI chatbots that they already use, asking them to provide Socratic questioning to help them learn. For example the following prompt works in typical major AI chatbots: “Use Socratic questioning to help me learn about <topic>.” See [another example](#) or try it yourself. The result is an individualized, dynamically generated learning session.

Further, the student can elevate control by asking AI to adjust the level of interaction (easier, harder, provide context, add keywords, provide search options, etc.). The learner can steer the thread of interaction and potentially focus on learning the material at hand without investing excessive time and energy on prompt creation. Learners can guide, and be guided by, their AI tutors, while benefiting from infinitely patient, customizing AIs. However, appropriate AI communication skills are required [3].

GOOD #2: Instructional Benefits

Artificial intelligence promises to increase teaching efficiency and effectiveness, while improving teachers’ environment and work experience. If teaching effectiveness is the amount a student learns for a given amount of effort, then teaching efficiency is the number of students taught times the effectiveness [4]. AI in this sense is a force-multiplier. To increase efficiency, teaching tasks may be delegated to AI which can act as both a surrogate lecturer, tutor and teaching assistant [5], customizing the pace and style of instruction to each individual student. However, students will need to learn how to interact well with other people too. Although there is some evidence that AI will be useful in teaching socialization [6], human interaction is also needed, so it is unlikely that instructors functioning in a collaborative academic environment will be completely replaceable.

AI is also useful in testing, grading and teaching materials development, which are for most instructors the least interesting facets of instruction. AI is especially useful here, although the quality of AI contribution is mixed. However, streamlining these facets may make significant contributions to instruction quality of life and the reduction of stress and burnout [7]. In terms of teaching materials development, AI may be useful in brainstorming [8]. Since AI tends to hallucinate, the brainstorming process has some attendant risks and requires a human-AI synergy for full value. That said, rough drafts of teaching materials, conversion of notes into slides, slides into readings, etc., are straightforward tasks for standard AI. In addition, educational computer games about a course topic can be created so quickly that an instructor can make them for students to play as homework problems. It is not necessary to know computer programming to use such tools (e.g., the artifact creator at <https://claude.ai/artifacts>). Commercial quality is not needed for it to be a useful [homework exercise](#).

In principle, AI automation may lead to time reallocation to more important aspects of teaching. Professional development for instructors regarding AI has been recognized and codified [9]. The

impact of AI on teaching is expected to deepen, making teaching more efficient and at the same time freeing instructors to focus on the most rewarding and human aspects of their profession.

GOOD #3: AI benefits for reading

The main benefits of AI to reading in higher education surround enhancing understanding of what is read. The most obvious way for AI to help is by answering a reader's questions about the material by way of a summary, explanation, elaboration, gist, quotations from the article, etc. An example prompt for this scenario is:

I have uploaded a dense paper, but you are an expert at explaining things to people who are unfamiliar with this paper. Please explain the sentence "<quote sentence or other passage here>". Be very brief and avoid a multi-paragraph text wall answer. Make it understandable to people who are unfamiliar with the topic. Include quotes from the paper where possible.

At that point, [a reader may follow up](#) with more specific questions about the sentence or passage at issue, or AI's explanation of it. This approach can also be applied recursively to dense AI-generated outputs.

This style of AI interaction can be applied by a student reading challenging classical material [10], a graduate student researcher reading a dense scholarly publication, or an established researcher reading a technical paper not in their area of particular expertise [11].

Meta-prompts may be added that address different reading levels, educational backgrounds, translations or [different languages](#) [12] [13]), or based on vocabulary [2]. Additionally, one might ask for an article to be dynamically distilled into segmented abstracts by length, focus or relevance.

While AI can flexibly and creatively provide information to support readers, it can also assist readers through active learning, such as by [pre-inserting questions](#) after each sentence or paragraph of an article, a simple yet comprehensive strategy. Alternatively, questions can be generated dynamically, as in the iRead system [14]. The dynamic approach can adjust as needed, steering readers toward deeper understanding with questions from different levels of Bloom's taxonomy [15].

Meta-prompting can direct an AI to use Socratic questioning instead of just providing information in response to reader questions. Standard AI chatbots readily comply, and can adjust the [Socratic Q&A process](#) to better match the reader's background and reading style upon request. The reader may hybridize Socratic and non-Socratic reading support by telling the chatbot when to switch between answering questions and asking them. Hybridizing still further, Socratic questioning can be one component of a more elaborate overarching reader support strategy; for details we simply let [AI speak for itself](#). The reading experience is changing rapidly, and the traditional method of simply reading the printed page will continue to recede.

GOOD #4: AI benefits for writing

AI can create fluid, readable text. Its understanding of what it writes is indeterminate. For good or ill, AI-generated text has become a mainstay of students in a hurry to submit written assignments of all kinds and has made serious inroads into journalism and professional writing. It has also been documented that research grant proposals drafted with AI help are more likely to receive funding [16].

Automated writing support predates AI with spell and grammar checking and enhanced editing functionality. AI takes this further by adding capability to suggest stylistic improvements for readability and to re-organize and re-purpose content. It also extends the ‘typeahead’ concept by [suggesting next words and phrases](#). A natural extension of this is to suggest subsequent sentences, paragraphs, and the like which seriously challenges the prevailing view of human authorship. In addition, AI may be used to analyze the literature [17], design experiments, write up the results for publication, and get 844 papers listed on an author’s Google Scholar profile for one year [18]. While generative AI cannot write great novels, at least not yet, its propagation throughout the writing ecosystem is here to stay and advancing apace.

Needless to say, the price for these advantages is the increased potential for abuse. The following four examples illustrate this point.

BAD #1: Fabricated literature reviews

Literature reviews have been the starting point for academic scholarship throughout most of recorded history. In computer science, two complementary journals, ACM’s Computing Surveys (1969-) and ACM Computing Reviews (1985-), are specifically dedicated to that role. In the last century, as science became bigger and more critical to the public welfare, global economics, and the global communication infrastructure, literature reviews became more sophisticated and detailed. George, et al. [19] identify several approaches to reviews that are found in modern publication venues:

1. exploratory reviews to provide a substantial contextual envelope for a subject
2. integrative reviews that provide a broader, diachronic contextual envelope with references to related subjects in different disciplines
3. meta-narrative reviews to achieve an in depth, inter-disciplinary contextual understanding of a topic
4. critical review that integrates opposing views to a subject
5. meta-analysis that contextualizes quantitative relationships between variables
6. meta-regression that establishes a concrete quantitative relationship between variables
7. mixed research synthesis that includes discussion of causal connections between variables or boundary conditions

While this study investigates review paradigms in public administration, the type and varieties of literature reviews described apply to a broader scope of scholarship. More detailed paradigms that are domain specific are also available [20].

These references circumscribe the type and variety of literature reviews that are relevant to peer-reviewed research publications. However, actual practice may be orthogonal to the best

practices. Literature reviews can be perfunctory. After all, a correct citation of a published resource neither implies that the resource was understood or, for that matter, that it was ever read. In some disciplines a FOLA (fear of leaving out) principle is a driving force. This motivates the scholar to be over- rather than under-inclusive. In big science and medicine for example, overlooking key references might lead to a rejection of the submission, or at least a time-consuming revision. And of course there are other unwholesome factors that may be in play such as the drive to include journal self-references, or a bias to feature critical citations by author/editor. With this in mind, we must ask how GenerativeAI would impact literature reviews.

One problem that has been identified is citation fabrication. Walters and Wilder document that this was a problem with both ChatGPT-3.5 and ChatGPT-4 [21]. They used the ChatGPT platforms to produce short literature reviews on 42 multidisciplinary topics and found that 55% and 18% of the resulting citations, respectively, were “fabricated and did not correspond to actual scholarly works” although they may have appeared to be deceptively real. These errors were attributed to GenAI hallucinations – i.e. factual errors which we have elsewhere subsumed under the descriptor GenAI bloviation [22]. In addition, 43% and 24%, respectively of the actual citations contained substantive citation errors and irregularities. Walters’ and Wilder’s article include references to over a dozen other articles that have reported similar results. Of course the key to unlocking the cause of fabricated citations and citation errors lies in an understanding how GenAI platforms work, which is impossible because they involve proprietary software and incompletely disclosed datasets. Since GenAI is opaque about functionality, it is an open question whether the source of these hallucinated citations lies in inconsistencies in the data or algorithmic inadequacies or improprieties. Whatever the cause, the GenAI output is unreliable and inadequate to the task of producing literature reviews. Gravel, et al. [23] describes the problem succinctly: GenAI is too often “confidently wrong” in its output. Walters and Wilder attribute this to its functionality: it is, at bottom, “fundamentally a text transformer – not an information retrieval system.” We have referred to this phenomenon as vacuous semantic omniscience [24]. In any event, these observations suggest fundamental flaws in the underlying design of GenAI, and not data processing errors.

A related activity, abstracting or gisting, is also being done with GenAI. Gao, et al. [30] found that GenAI derives “believable” scientific abstracts from medical research publications, notwithstanding some failures in format. Domain specialized reviewers misclassified approximately one-third of the GenAI abstracts as original even though they were aware that some of the elements of the set of abstracts were seeded by GenAI – an advantage not accruing to an unwary reader. The authors freely admit that “Given its ability to generate abstracts with believable numbers, [GenAI] could be used to entirely falsify research.”

BAD #2: Misrepresentation of GenAI output as your own work

A second example of unscholarly behavior is the most common of all: misrepresentation of GenAI output as your own work. This is not plagiarism in the strictest sense of the word since the document was not produced by a person. Some universities and publishers are reluctant to designate GenAI users as “authors” for this reason. In its recent GenAI acceptable use policy, Princeton resists the temptation unilaterally treat GenAI output as a source document: “Generative AI is not a source... because the output is not created by a person. If generative AI

is permitted by the instructor, students must disclose its use rather than cite or acknowledge the use, since it is an algorithm rather than a source.” (<https://rrr.princeton.edu/students-and-university/24-academic-regulations>, section 2.4.7). We’ll concede this issue to Princeton, and subsume the undisclosed submission of GenAI output as your own work under an additional category, misrepresentation.

We illustrate the potential problem of misrepresentation by reference to one study of its occurrence in universities. Yeardon, et al. [26] showed that GenAI is effective at producing high-quality university physics essays. This study used GenAI to produce short-form (300-word) physics essays. Human domain experts evaluated these essays and determined that as a group they would receive high marks (First Class in UK) which suggests that they “represent a significant threat to the fidelity of short-form essays as an assessment method in Physics courses.” Brainard has documented that scholarly journals have been struggling with the problem of misrepresentation for some time [27]. We agree with Carlson and Laff that a large part of this problem is that universities have wandered too far away from the primary goal of providing a diversified, well-rounded education [28]. Of course the prevailing publish-or-perish paradigm widely used in faculty assessment may be expected to exacerbate this problem

BAD #3: Plagiarism

In addition to specious literature reviews and misrepresentation, GenAI also automates another unscholarly behavior: plagiarism. We admit that the difference between plagiarism and misrepresentation of authorship is imprecise, but the absence of a dichotomy does not imply the absence of a distinction. One may think of intentional plagiarism as deliberate misrepresentation that is done intentionally and for personal gain – whether for commercial gain or for purposes of self-promotion, advancement, economic advantage, educational advancement, prestige, etc. As is the case with misrepresentation, GenAI-assisted plagiarism is widespread, subtle, and very difficult to detect and rears its ugly head in several venues, including education, the professions, business, politics, and perhaps most concerning, scholarly publication. A generic problem with GenAI based on proprietary code is that it is exceedingly difficult to determine LLM-harvested content from output, even algorithmically, without comparing the output with every content source touched by the LLM in training.

As Chris Westfall recently put it, “the plagiarism potential [of GenAI] is potent.” [29] Some recent research that showed that ChatGPT performed at or near the passing threshold for all three tests that make up the United States Medical Licensing Exam without any specialized training [30]. The research report concludes that “Our study suggests that large language models such as ChatGPT may potentially assist human learners in a medical education setting, as a prelude to future integration into clinical decision-making.” In this setting, the thought occurs: from what sources and authors did the answers provided on the licensing exam question derive? And to what extent does the individual potential licensee wish to claim ownership and responsibility for the answers? Assuming that the answers are correct, they came from some authoritative source(s). In a very intuitive sense, these source(s) were plagiarized – re-purposing the content does not break the chain of authorship. We set aside discussion of the separate issue of whether society should be willing to accept that vetted physicians merely ‘perform at or near a passing

threshold' on these exams. In any event, it must be admitted that these research results send out alarming signals.

In each of these venues, GenAI is seen to amplify pre-existing problems. Throughout history people have sought to take credit for the accomplishments of others – so misrepresentation and plagiarism is nothing new. But GenAI makes the process both effortless and anonymous, and this justifies special attention.

BAD #4: Refinement and fabrication of fake content

We conclude this brief overview with fourth example of GenAI abuse in education: fabrication and the refinement of fake content. While bogus or fake content is not particular to GenAI output, it holds considerable potential for refining the content and making it both harder to identify and more palatable to general audiences. Bauer and von Hohenberg showed, prior to the widespread popularity of GenAI, online fake sources had already begun to compete effectively with veridical content for popular attention [31], and Rhodes showed how the social media ecosystems can multiply this effect [32]. Taken together, social media platforms can amplify bogus, fraudulent, GenAI content.

The term fabrication has many denotations from an act of manufacturing something to a deliberate act of deception. In the case of GenAI we focus on one specific meaning: fabrication as the act of creating a literary or artistic artifact that artificially represents a state of affairs without any human cognitive investment. We take no position on whether any deception was involved in the creation of the artifact, or whether the resulting artifact is a truthful or factual representation of anything. In this way, we avoid entering into discussion of political and social fabrications, intentional lies, deep fakes, and other sundry social and psychological distractions not specific to higher education. We seek to emphasize the distinction between a fabrication and the product of human ratiocination. As a primitive example, the selection of the one syntactically and semantically well-formed story taken from myriad tokens that resulted from the random cutting and pasting words cut from newspapers by some automated process would constitute a pure fabrication, whereas a story written by a competent native language speaker would not.

Examples of such abuse of GenAI in scholarship abound. Our four examples neither exhaust the range of bad uses of GenAI nor necessarily represent the most socially reprehensible instances. But if society is to minimize the deleterious effects of this abuse, it must closely monitor the observed patterns and trends. And this monitorial effort cannot be left to automation. At this point, a competition is emerging between the developers of GenAI platforms and GenAI detectors, with the overwhelming advantage accruing to the GenAI platforms for both economic and technical reasons. We emphasize that we are not opposing the use of GenAI in education, medical research and science, but that we are offering a caution that much more clarity needs to be offered regarding the difference between enhancing our cognitive capacities and substituting for them. Society has as yet to draw clear guidelines between using the output of GenAI as a tool used to assist a scholarly effort on the one hand, and relying on the output as the substitute product on the other. As we have pointed out, university approaches to acceptable use policies for GenAI differ widely, and are nowhere near approaching a consensus [33].

As an aside, on June 23, 2025 the U.S. District Court for the Northern District of California ruled in *Bartz, et al. v. Anthropic* that harvesting copyrighted materials for LLM training is justified under the “fair use” provision of copyright law [34]. Far-reaching implications relating to fabrication follow from this ruling, although the details of these implications won’t be clear for several years. Will society come to accept the Bartz decision over time, or reject the notion that the use of pirated content by LLM training (or anything else, for that matter) is legal (or ethical)? This also brings into question whether the Bartz decision is consistent with the view that all GenAI output should be put into the public domain.

To this point, we’ve tried to be balanced in our illustrations of the use of GenAI in education. As we’ve indicated, the use of GenAI in education and for scholarship is a mixed bag. If one might take away a single message from this discussion it would be that it may be sensible to encourage GenAI use whenever the output of such use is supplementary and incidental to the final product of the effort. The use of GenAI to suggest topics, themes, perspectives, identifying errors, jog memory, factcheck, search, debug, document, etc. for a research paper seems to us to be an entirely reasonable use of the platforms. However, we would argue that the use of GenAI to produce a claimed-authorship finished product is controversial at the least and may be viewed by many as improper and unprofessional. In saying this, we admit to a huge gray area between GenAI as a support tool and GenAI as a surrogate author. Social scientists will be weighing in on this distinction for years.

We’ve focused on what we might call ‘native’ advantages and disadvantages of GenAI – those relating to the actual use of GenAI in education. There are, however, some serious negative externalities that may result which are not directly connected to the actual use but are rather psychological, social and cultural byproducts. We categorize some of these under the rubric of “ugly”.

UGLY #1: Composing Essays and Computer Programming

There are some aspects of GenAI use that are – or should be – considered alarming by everyone. We begin with some recent studies that suggest that the use of GenAI for essay writing reduces the cognitive load significantly enough to interfere with cognitive engagement in a task [35]. In one recent study, when subjects were grouped by use of support tools {search engine, GenAI, and none/brain only}, electroencephalography (EEG) showed that the three groups demonstrated “significantly different neural connectivity patterns, reflecting divergent cognitive strategies.” GenAI subjects “showed weaker neural connectivity and under-engagement of alpha and beta networks.” The authors suggest a likely decrease in learning skills and under-performance at all levels: neural, Linguistic, and scoring. Thus, there is some evidence that the use of GenAI for scholarship can be toxic [36].

Other studies that focus on university students’ use of GenAI for programming assignments are equally alarming. For example, Lepp and Kaimre discovered a negative correlation between the frequency of use of GenAI tools and student performance [37]. This suggests that the tools themselves may have a negative impact on learning programming. Similar results were found in a recent study conducted by Eric Klopfer at MIT. Three groups of students in a programming class were given a programming assignment in Fortran. One group was allowed to use ChatGPT,

another Meta's Llama, and the third could only use a search engine. The group results ranked ordered by speed were (1) ChatGPT, (2) Llama, and (3) search engine, so the use of GenAI definitely increased throughput. But, subsequent testing of the students on the underlying meaning of the solution revealed a very different result. None of the ChatGPT students passed, only half of the Llama students passed, and all of the search engine students passed. Klopfer summarized his findings: "This is an important educational lesson....working hard and struggling is actually an important way of learning. When you're given an answer, you're not struggling and you're not learning. And when you get more of a complex problem, it's tedious to go back to the beginning of a large language model and troubleshoot it and integrate it." [38] These essay and programming examples provide some evidence that any productivity advantage that might accrue to the user from GenAI may come at the expense of knowledge and understanding. To put a finer point on this, a difference must be drawn between a conversational and informal use of a tool (whether natural or automated) on the one hand, and the scholarly pursuit of knowledge on the other. There is no shortage of research studies that report similar results. Further, many STEM educators (ourselves included) have witnessed changes over the past few years in student outcomes assessments that are consistent with these results [39]. We have observed what appears to be an inverse relationship between student reliance on GenAI for student assignments and subsequent exam performance over the same material. We see this as a form of cyberdilettantism, characterized by lowered cognitive investment and minimal cognitive inertia [40]. Educators who find these results disturbing and cautionary would be well served to be proactive in providing detailed course guidelines on GenAI use. The importance of this suggestion has been confirmed by a recent report that GenAI has already found its way into the Canadian Computing Competition which has disqualified some competitors [41].

UGLY #2: Labor Arbitraging

Our second example deals with economic implications of GenAI, in particular in terms of labor economics. GenAI is uniquely positioned to be disruptive in the workplace. As such it has much the same potential for wage deflation as the H1B visa – which, accordingly, is a primary reason why it is so highly touted by investors and corporations. Why pay higher salaries for sought-after skills when acceptable substitutes are available? In terms of classical economics, when limiting consideration to labor cost as the only variable, labor arbitraging - whether by employing people who are willing to work for lower salaries or using GenAI to eliminate jobs entirely – provides absolute advantage for the employer. Of course, from a societal perspective, labor arbitraging is zero-sum: whatever financial advantage accrues to the employer is proportional to a consequential financial disadvantage to the labor market. Of course, the problem with this approach is that economic interaction is too complex for an analysis with only one variable, and as a result the absolute advantage is illusory and cannot be sustained as it fails to take into account other critical factors such as the potential of the labor alternatives for innovation and discovery, process improvement, supportive adhocracy, market and social awareness, the potential to avoid institutional instability and conflict, etc. The somewhat analogous situation in the role of tariffs in trade policy, led to the criticism of mercantile capitalism. Whatever immediate advantages of labor arbitraging and mercantilism tend to dissipate over time under the lens of more complete and thorough analyses of economic productivity and efficiency. In the academic arena, the labor arbitraging is likely to take early root in service and support roles such as grading; reporting; evaluating students, faculty and staff; planning and scheduling,

and so forth. As former administrators, it is hard for us to imagine how SWOT analyses or annual faculty evaluations could be responsibly handled by GenAI. For example, how might one guarantee that the training of the LLM was done on accurate, relevant data? And as any teacher will attest, grading all but the simplest true-false and multiple choice exams and most mundane homework is nuanced, and nuance falls outside the purview of GenAI.

UGLY #3: GenAI may lead to inflated self-efficacy

A third unfortunate effect of the use of GenAI is psychological: inflated self-efficacy may be a byproduct of sustained use. Self-efficacy is not a sufficient condition for academic or scholarly ability. Self-deception may also be at work. Self-efficacy must be conditioned by internal and external feedback. Were one to see that certain patterns of behavior continually return high marks on exams, positive recognition from knowledgeable, respected peers, continued success in the exercise of skills, etc. one might legitimately assume some degree of self-efficacy. But, GenAI eliminates the continuous feedback from stakeholders which, in turn, may lead to ‘irrational exuberance’ and delusion. The process by means of which legitimate self-efficacy is achieved, the experienced mastery if you will, circumscribes a general-purpose, interactive learning system with multi-sourced and many-directional information stimuli, memory, a cognitive framework, feedback mechanisms, recognizers and analyzers of verbal and non-verbal patterns, and so forth. This is what Robin Milner called ‘conceptual armoury’ [40]. Our point is that irrational inflation of efficacy expectation is likely to lead to undesirable social consequences, and the resulting behavior may well be accompanied by negative externalities. GenAI is transformative in this regard. It appears to be an ideal breeding ground for such irrational inflation of self-efficacy.

UGLY #4: GenAI may lead to cognitive decline

Our fourth consideration is the possibility that sustained use of GenAI can lead to cognitive decline. Unlike other online digital tools such as wikis, encyclopedias, dictionaries, search engines, and the like, GenAI provides a dynamic, immersive environment by simulating human conversation. There is emerging evidence that suggests that this new GenAI online paradigm has the potential to become a substitute for human interaction that could potentially lead to cognitive atrophy. Dergaa, et al. [42] refer to this as artificial-intelligence-chatbots-induced cognitive atrophy (AICICA). This suggestion is consistent with the MIT research reported earlier on essay writing [34], a trend actually predicted by GenAI itself [43]. This trend, which has been associated with the practice of cognitive offloading, has been shown to negatively correlate with critical thinking [44]. One recent study of cognitive offloading has found an interesting twist on the Dunning-Kruger Effect (DKE) which holds that low-performers on cognitive tests tend to overestimate their abilities. With the use of GenAI, DKE doesn’t hold: higher AI literacy seems to breed overconfidence [45]. Though unintuitive, this is consistent with our observations about inflated self-efficacy, above.

Conclusion

As we mentioned above, our categorization reflects our experience. We would be remiss were we to fail to mention that other categorizations may be equally useful. For example, some might

suggest the inclusion of the use of GenAI to stimulate ideation or creativity as good [46]. Or the fact that the use of automation in higher education can be a distraction as a bad [47]. We might include as an exemplar of the ugly side of GenAI use that there is compelling evidence that GenAI encourages such unethical behavior as cheating [48] and future releases may have serious security implications [49]. Each of these could also qualify as reasonable inclusions in a modest holistic perspective. In fact, a comprehensive survey of the good, bad, and ugly aspects of the use of GenAI in higher education would be welcome. We shall restrict ourselves to this primer.

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